Physics-Informed Parameter Identification for Zig-Zag Pneumatic Soft Bending Actuators Using a Reduced-Order Continuum Model

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ELEC 395 Final Project

***Abstract*—Soft pneumatic bending actuators are widely used in compliant grippers, but their force and deformation responses depend strongly on pressure, geometry, and effective structural stiffness. High-fidelity finite element models are informative but expensive, while purely data-driven regressors often generalize poorly under sparse measurements. This project studies a physics-informed neural network (PINN) for inverse identification of a zig-zag pneumatic soft actuator using a reduced-order continuum model. The actuator is represented by an inextensible planar centerline with effective bending stiffness EI_{eff} and pressure-to-actuation coefficient k_p. Synthetic benchmark data are generated for a 200 mm long actuator with a 22×22 mm cross-section, 2 mm wall thickness, and a zig-zag compliance factor. The PINN jointly predicts centerline shape and identifies EI_{eff} and k_p from sparse shape, tip-displacement, and blocked-force observations, and is compared against a baseline multilayer perceptron (MLP). Results show that the PINN identifies k_p accurately and reproduces the blocked-force response well, while still underestimating large-deformation tip displacement at high pressure. Ablation studies further show that blocked-force supervision is essential for parameter identifiability and that the PINN is more data-efficient than the MLP in sparse-data regimes.**

I. Introduction

Soft pneumatic actuators are widely used in soft robotics because they provide compliant interaction and adaptable gripping, but their modeling and calibration remain difficult owing to geometric nonlinearity, material nonlinearity, and actuation complexity [1], [2]. Reduced-order models are attractive because they can lower computational cost compared with repeated high-fidelity simulation while retaining actuator-level interpretability [3]. In parallel, physics-informed neural networks (PINNs) have emerged as a useful scientific machine learning framework for forward and inverse problems because they combine sparse data with governing equations and boundary conditions [4], [5].

This project asks two questions: (i) can a PINN identify physically meaningful effective parameters from sparse observations, and (ii) does blocked-force supervision improve identifiability and prediction quality relative to using shape and tip displacement alone? A baseline MLP is used to assess the benefit of the physics-informed formulation.

II. Reduced-Order Model and Inverse PINN

The actuator centerline is parameterized by arc coordinate $s \in [0, L]$ with planar state variables $(x(s), y(s), \theta(s))$. Under an inextensible-centerline assumption,

$$\frac{dx}{ds} = \cos\theta, \qquad \frac{dy}{ds} = \sin\theta. \tag{1}$$

Pressure loading is represented by an effective moment balance

$$EI_{\text{eff}}\,\theta''(s) + k_p\,p = 0, \tag{2}$$

with clamped-free boundary conditions

$$x(0) = 0, \quad y(0) = 0, \quad \theta(0) = 0, \quad \theta'(L) = 0. \tag{3}$$

The benchmark actuator is a simplified zig-zag pneumatic bending actuator with total length $L = 200$ mm, outer cross-section 22×22 mm, wall thickness 2 mm, and a zig-zag compliance factor of 0.85. The active zig-zag region is 110 mm long, preceded and followed by 20 mm plain regions. The benchmark pressure range is 10–100 kPa.

Blocked force is defined as the tip reaction force required to hold $y(L) = 0$ under pressure loading. In the benchmark data, it is computed using a constrained boundary-value solver and used as a supervised observation.

The PINN takes (s, p) as input and outputs $(\hat{x}, \hat{y}, \hat{\theta})$. The trainable physical parameters are EI_{eff} and k_p. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{phys}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{shape}} + \mathcal{L}_{\text{tip}} + \mathcal{L}_{\text{block}}. \tag{4}$$

A purely data-driven MLP with the same input-output variables is trained as a baseline.

III. Data Generation and Experiments

Synthetic benchmark data are generated from the reduced-order solver over pressures from 10 to 100 kPa. Train/validation/test splits are defined over pressure levels. Evaluation metrics include RMSE for centerline coordinates, tip displacement, and blocked force, together with percentage error for identified EI_{eff} and k_p.

Two additional experiments are performed. Experiment 1 compares PINN training with and without blocked-force loss. Experiment 2 compares PINN and MLP under low-, medium-, and high-data regimes to assess sample efficiency.

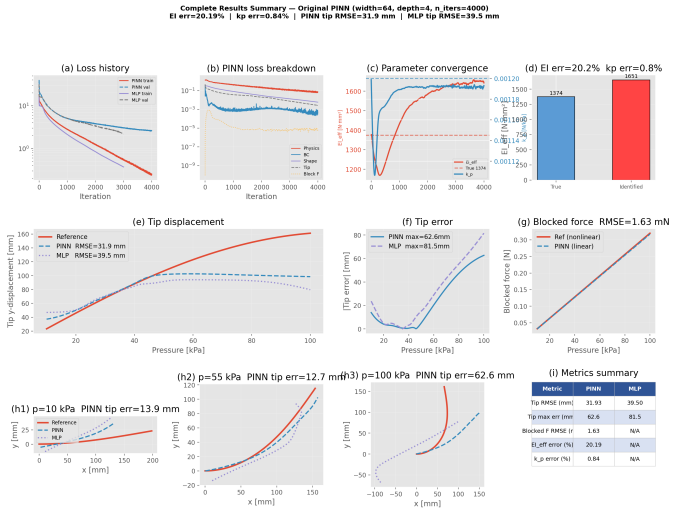

Fig. 1. Representative summary of baseline training, parameter identification, pressure-response curves, and centerline comparisons.

IV. Results

In the baseline setting, the PINN identifies the actuation coefficient accurately: k_p error is 0.84%, and the blocked-force RMSE is 1.63 mN. The effective bending stiffness remains harder to identify, with approximately 20% error, indicating that blocked-force observations constrain k_p much more strongly than shape observations constrain EI_{eff}. For tip displacement, the PINN outperforms the MLP, with RMSE of 31.9 mm versus 39.5 mm, but both models underestimate the response in the high-pressure regime where large deformation becomes important.

Experiment 1 shows that blocked-force supervision is essential. Removing the blocked-force loss increases the k_p error from 0.84% to 40.92% and the blocked-force RMSE from 1.63 mN to 79.78 mN, while also worsening tip-displacement RMSE from 32.0 mm to 39.2 mm. This confirms that blocked-force observations resolve a major EI–k_p identifiability degeneracy.

Experiment 2 shows that the PINN is consistently more data-efficient than the MLP. In the low-data regime, the PINN achieves 42.4 mm tip RMSE versus 59.8 mm for the MLP, a 29% reduction, and improves shape RMSE from 47.3 mm to 36.8 mm. Similar trends hold in the medium- and high-data regimes, with lower mean error and lower variance across seeds for the PINN.

V. Discussion and Limitations

These results suggest that the current PINN formulation is already useful for identifying actuation-related parameters and force response, especially when blocked-force measurements are available. The main limitation is the simplified mechanics model: the zig-zag chamber architecture is represented only through an effective compliance factor, and the large-deformation response remains under-modeled. As a result, the inverse model fits blocked-force and k_p well but still underestimates high-pressure tip displacement and compensates part of the shape mismatch by biasing EI_{eff}. The next logical step is to calibrate the reduced-order model against 2-D or 3-D FEM data for a more realistic zig-zag chamber geometry and then repeat the inverse PINN study.

VI. Conclusion

A PINN-based inverse framework for zig-zag pneumatic soft actuators was developed and evaluated on synthetic benchmark data. The method accurately identified the pressure-to-actuation coefficient and reproduced the blocked-force response, while outperforming a baseline MLP in sparse-data regimes. The main remaining challenge is accurate recovery of large-deformation centerline shape and tip displacement, which points to the need for richer reduced-order physics or FEM-grounded calibration. Overall, the study supports PINNs as a promising tool for actuator-level parameter identification in soft robotics.

References

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